

Integrating the agent-based approach with hydraulic analysis of the water distribution network to a realistic micro-scale simulation of end-users in GAZ City, Iran

Zohre Fakhari^a, Azadeh Ahmadi^{b,*} 

^a Dept. of Civil Engineering, Isfahan University of Technology, Isfahan, Iran

^b Faculty of Civil, Water and Environmental Engineering, Shahid Beheshti University, Tehran, Iran

ARTICLE INFO

Keywords:

Agent-based simulation modeling
Water conservation behavior
Water demand management
Complex phenomena
WNTR
Autonomous agents

ABSTRACT

Study region: GAZ City water distribution network in Isfahan province, Iran

Study focus: In conventional water distribution network models, consumer water demand and water use behaviors are often oversimplified. These models are typically one-directional, simulating how water distribution networks' performance affects nodal demand estimation, neglecting complexity behind mutual interactions between household and water distribution network behaviors. This study proposes a novel framework, integrating an agent-based model with a hydraulic simulator, to simulate realistic consumer water use, which can capture reciprocal dynamics between end-user behaviors and the hydraulic performance of water distribution networks. These dynamics simulate how network pressure influences individual water use and how consumer water use feeds back hydraulic performance. The primary objective is to simulate these two-way interactions under abnormal conditions, thereby strengthening hydraulic analysis and supporting the assessment of social satisfaction.

New hydrological insights for the region: In the proposed framework, individual water-use behaviors across different activities are simulated under varying pressure conditions, accounting for both individual characteristics and social interactions. Results showed that consumer dissatisfaction was concentrated during peak demand periods under low-pressure conditions, while water use exceeded demand during off-peak times. Furthermore, modifying consumer attitudes toward water conservation improved network performance by 13 % without altering hydraulic parameters. By simulating realistic consumer responses, the framework provides a tool for evaluating the outcomes of water management policies under abnormal conditions.

1. Introduction

In the hydraulic analysis of water distribution networks (WDNs), nodal demand and nodal outflow are defined as the water needs and withdrawals of nodal consumers, respectively (Gleick, 2003). Based on the procedure used to calculate nodal outflow, the hydraulic analysis methods for WDNs are categorized into different approaches. In the demand-driven analysis (DDA) approach, it is assumed that nodal demand is always fully satisfied, regardless of the effect of pressure on nodal outflow. Thus, under this analytical approach, nodal demand and nodal outflow are considered identical. In some studies, pressure constraints are incorporated alongside

* Corresponding author.

E-mail addresses: z.fakhari@cv.iut.ac.ir (Z. Fakhari), az_ahmadi@sbu.ac.ir (A. Ahmadi).

<https://doi.org/10.1016/j.ejrh.2025.102805>

Received 22 May 2025; Received in revised form 22 September 2025; Accepted 23 September 2025

Available online 29 September 2025

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DDA for effective pressure management in WDNs (Kakeshpour et al., 2024a) and the design of district metered areas (DMAs) (Kakeshpour et al., 2024b). However, the results of DDA are valid only under normal operating conditions and during the design phase of WDNs, and do not produce realistic results during abnormal operating conditions (Tabesh et al., 2014).

Abnormal operating conditions refer to pressure deficits in the WDN, resulting from three main factors: (1) an imbalance between water resources and demand (e.g., population growth, climate change) (Sattler et al., 2023), (2) hydraulic factors (e.g., inefficient pressure management, demand exceeding WDN capacity, increased pipe hydraulic resistance or leakages) (Giustolisi and Walski, 2012), and (3) structural factors (e.g., pipe bursts, pump failures) (Giustolisi and Walski, 2012). Structural failures, such as pipe bursts and pump failures, occur infrequently. In contrast, abnormal operating conditions related to hydraulic and supply factors are relatively common in some WDNs and frequently expose consumers to service disruptions. During service disruptions, pressure deficiencies may result in demand not being fully satisfied.

Under abnormal operating conditions, the pressure-driven analysis (PDA) approach is employed (Pathirana, 2010; He et al., 2015; Rossman et al., 2020). In this approach, nodal outflow is modeled as a function of both nodal demand and available pressure, supported by head-outflow relationships (HORs). Thus, unlike DDA, under this analytical approach, nodal demand and nodal outflow are not considered identical. Numerous HORs have been proposed; among these, Wagner relationship (Wagner et al., 1998) is the most commonly used in PDA due to its ability to produce more realistic results (Shirzad et al., 2013; Chang et al., 2019). Nevertheless, the Wagner relationship has certain limitations, as identified in previous studies, including the neglect of the temporal and spatial distribution of nodal demand and variations in water use behavior (Burkhardt et al., 2023; Schück and Lansey, 2018). Capturing the diversity of consumer water use behavior in hydraulic analyses of WDNs remains challenging, as behavior varies between individuals and may change for an individual depending on circumstances (Díaz et al., 2024).

In recent years, researchers have modeled the complexity and diversity of human behavior using agent-based models (ABMs). In ABMs, complex behavioral patterns and emergent phenomena are simulated by defining independent agents with varying characteristics and behavioral rules. In these models, each agent assesses its state, makes decisions based on predetermined rules (Bonabeau, 2002), and updates its behavior in response to environmental factors and interactions with other agents (Ramsey and Berglund, 2021). ABMs have been used to simulate the diverse water demand behavior of consumers in response to water demand management policies and varying hydrological conditions (Athanasiadis and Mitkas, 2005; Chu et al., 2009; Koutiva and Makropoulos, 2016, 2019; Ali et al., 2017). Furthermore, ABMs have been employed to model consumer behavior in adopting water-saving devices and water conservation behaviors (Rasoulkhani et al., 2018; Shahangian et al., 2023). In these studies, the consumer community is modeled using ABM without addressing the interaction between consumer behavior and the hydraulics of WDNs.

Integrated ABMs and WDN hydraulic simulators have been used to evaluate the performance of water infrastructure under various operating conditions (Liu et al., 2016), the adoption of water reuse technologies (Kandiah et al., 2013, 2014) and water-efficient devices (Tourigny and Filion, 2018), the consequences of demand management policies (Alghamdi et al., 2024), and crisis scenarios (Zechman, 2007; Shafiee and Berglund, 2017; Kadinski et al., 2022; Vizanko et al., 2024, 2025).

In these studies, ABMs have been integrated with hydraulic simulators to evaluate the effects of demand variations on the quantitative and qualitative parameters of WDNs. Some of these studies have also investigated the feedback effects of WDN simulation outcomes on consumer behavior. For example, in the study by Kandiah et al. (2014), the strategies of the management agent were updated based on demand within the WDN and the water reuse system, aiming to encourage consumer adoption of reuse technologies. Moreover, researchers such as Shafiee and Berglund (2017), Kadinsky et al. (2022), and Vizanko et al. (2025) have simulated the bidirectional interactions between water quality within the WDN and consumer demand. These previous studies have coupled ABMs with WDN hydraulic simulators to model the bidirectional interactions between water quality and consumer demand at the individual or activity level. However, these frameworks neglect micro-scale water use behaviors that are directly influenced by hydraulic factors such as pressure. This neglect stems primarily from two factors. The first is the common assumption that water use equals demand whenever sufficient pressure is available throughout the WDN. The second is the reliance on HORs that represent the effect of pressure on water use in an aggregated form at the node level, rather than capturing variations at the micro scale. As a result, a notable gap remains in the literature regarding the explicit modeling of dynamic interactions between WDN pressure and consumer water use behavior at the micro scale.

In this study, an innovative WDN hydraulic analysis framework is presented that employs ABM as an alternative to HORs, enabling the effective incorporation of diverse consumer behaviors into nodal outflow estimation. The framework integrates three interconnected sub-models: a WDN hydraulic simulator to model water system dynamics, a water demand simulator to generate water demand components, and a water use simulator to capture the water use behavior of consumers. In this framework, consumers are modeled as independent decision-makers within the ABM, with their behaviors determined by individual characteristics and social interactions. A key novelty of the proposed approach is the iterative coupling of the WDN hydraulic simulator and water use simulators, enabling dynamic feedback between WDN hydraulics and consumer decisions. This framework enhances hydraulic analysis accuracy and provides utilities with actionable insights into consumer satisfaction during low-pressure scenarios.

The framework process is briefly presented in the following section, followed by a detailed explanation of the sub-models. Subsequently, the framework is applied to the WDN of Gaz City, and the results are discussed and compared with those obtained from PDA. Before concluding, the results from scenarios involving consumer water conservation attitude and reservoir head variations are presented.

2. Methodology

2.1. The overall design of the study

The proposed framework consists of three sub-models, each developed within this study: 1) the hydraulic simulator sub-model, which simulates the hydraulics of the WDN using the Water Network Tool for Resilience (WNTR) package, 2) the water demand simulator, implemented as an ABM to represent consumer demand, 3) the water use behavior simulator, also developed within the agent-based modeling framework to replicate consumer water use behaviors. The ABM components were developed using the NetLogo software platform. Subsequently, integration of all three sub-models was performed within the Python environment, facilitating dynamic bidirectional interactions between consumer behavior and the hydraulic conditions of the WDN. A flowchart illustrating the overall research framework is provided in Fig. 1. The overall design of the proposed framework, as illustrated in the flowchart, is presented here, and the methodology for each sub-model is described in the subsequent sections.

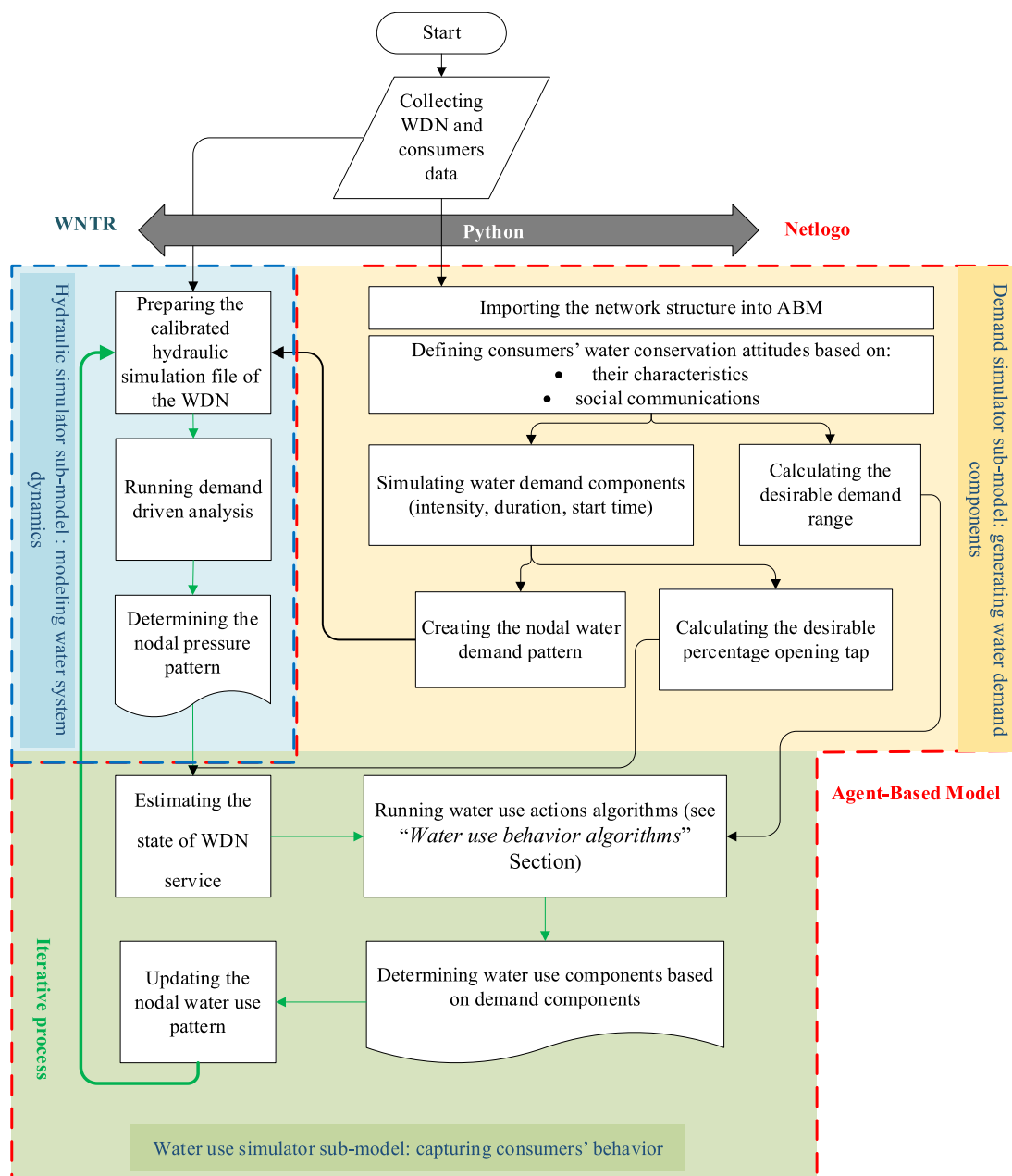


Fig. 1. The structure of the present framework.

Initially, in the water demand simulator sub-model (orange section of the flowchart), the WDN structure was incorporated into the ABM. The consumer community was defined at the WDN nodes, and their characteristics and social interactions were assigned. Subsequently, their water conservation attitudes were allocated based on their characteristics and the effect of social interactions. Next, the demand components (including the intensity, duration, and start time) for various consumer water use activities were determined. Following this, a desirable demand range was defined for each iteration of water-use activities based on consumers' demand intensity and water conservation attitudes. In addition, the "nodal water demand patterns" were generated based on demand components and then incorporated into the hydraulic simulator sub-model (blue section of the flowchart). By coupling the WDN hydraulic simulator and water demand simulator sub-models, the water system was analyzed to calculate the available nodal pressure.

The nodal pressure pattern was input to the water use simulator sub-model (green section of the flowchart). In the water use simulator sub-model, the state of WDN service was determined based on the nodal pressure pattern and then input into the water use behavior algorithms. In the water use behavior algorithms, the water use components (i.e., the updated water demand components based on consumer actions) were determined based on these two inputs: the desired demand range and the state of WDN service. Subsequently, by coupling the water use simulator sub-model and WDN hydraulic simulator sub-model, the hydraulics of the WDN was reanalyzed based on the nodal water use pattern. If the new nodal pressure patterns differed significantly from the previous ones, consumers' water use components were redetermined based on the updated water pressure pattern. This process was repeated until no significant difference was observed between the pressure patterns of the two stages. Convergence between the nodal pressure patterns in the two stages was measured using the Euclidean norm criterion.

2.2. Hydraulic simulator sub-model

In the developed framework, the hydraulics of the WDN was analyzed based on consumers' water use components, and the water use components were influenced by available nodal pressure. To capture this two-way relationship between the hydraulics of the WDN and consumers' water use behavior, the hydraulic simulation sub-model was coupled with the water use simulator sub-model in an iterative process. In each iteration, the hydraulic analysis sub-model executed an extended period of DDA on a minute time scale over 24 h. The hydraulic performance of the WDN was evaluated using the demand satisfaction ratio, defined in Eq. (1), which was adapted from the study by Hayuti et al. (2008).

$$\text{Demand satisfaction ratio} = \frac{\text{Network water use}}{\text{Network water demand}} \quad (1)$$

The demand satisfaction ratio denotes the proportion of water use in the WDN relative to the corresponding demand. This ratio can be less than, equal to, or greater than one. A value below one indicates that the water use volume within the WDN is lower than the total demand, a value of one signifies an exact match between water use and demand, and a value above one reflects a water use volume exceeding the total demand. For the hydraulic simulation of the WDN, the open-source EPANET software's programming interface was employed within the Python environment, referred to as WNTR (Klise et al., 2020), which had been used in previous studies to simulate unusual scenarios (Moriarty et al., 2016; Antonowicz et al., 2020).

2.3. Agent-based behavioral simulation sub-models

The diversity of water demand and water use behaviors of consumers is key benefits of the model. The hydraulics of WDNs are closely related to water demand and the water use behavior of consumers on spatial and temporal scales. ABMs have the ability to capture the behavior of each consumer as an independent decision-maker with different properties (Wilensky and Rand, 2015). Therefore, in this study, the mutual interaction between WDNs and social components from the perspective of water demand and use was modeled through the ABM coupling with the hydraulic analysis tool. Among the various tools for implementing the agent-based concept, the NetLogo software is suitable for relatively large-scale models and provides simpler modeling capabilities (Abar et al., 2017). The presence of an extensive library and an active support community are other advantages of this application (Robertson, 2005; Nikolai and Madey, 2009). Furthermore, the NetLogo and Python environments are connected using the PyNetLogo tool (Jaxa-Rozen and Kwakkel, 2018). Accordingly, in this study, the ABM was implemented using the NetLogo 6.2.0 software. The details of different parts of the ABM, including the water demand simulator sub-model and the water use simulator sub-model, are explained below. The full Overview, Design concepts, and Details + Decision (ODD+D) protocol (Müller et al., 2013) of the model is also provided in the [Supplementary Material](#).

2.3.1. Water demand simulator sub-model

2.3.1.1. Consumers and characteristics. Water pressure varies spatially across nodes in the WDN, causing consumers to experience different levels of service. Since the state of WDN service for consumers depends on their position within the network, it is necessary to spatially assign agents to WDN nodes. Therefore, the algorithm developed by Hajebi et al. (2013) was employed to import the WDN structure into the agent-based environment. Subsequently, the covered population was modeled at the nodes in the form of individual agents grouped into households with various socio-demographic characteristics.

In each household, several properties were assigned, including the gender and age of the household head, income, dwelling size, ownership status (owner or tenant), dwelling type (apartment or non-apartment), floor number, and whether the water meter is shared in apartment buildings. Additionally, five types of taps with specific elevations were defined for each household, including kitchen,

outdoor, WC, shower, and bathroom taps. Furthermore, some properties were allocated to each individual, including age, gender, level of education, employment status, as well as two cognitive beliefs: environmental concerns and drought beliefs.

The water conservation attitude of consumers was determined based on several properties, including gender, age, level of education, household size, gender of the household head, dwelling size, income, whether the water meter was shared in apartment buildings, environmental concerns, and drought beliefs (Harlan et al., 2009; Willis et al., 2013; Koutiva and Makropoulos, 2016; Rasoulkhani et al., 2018). First, a grade between one to five was assigned to each property of the agents. In the second step, the attitude rank of each agent was determined by summing the attribute grades. Since each consumer had ten properties graded from one to five, the attitude ranks ranged from 10 to 50. In the third step, the attitude rank was normalized, and the normalized attitude rank (NAR) parameter was expressed as a number between zero and one, representing the level of water conservation attitude of consumers.

2.3.1.2. Social communications and impact. In the social component, consumers communicate with each other, and their water conservation attitude was influenced by the other consumers with whom they were in contact. In this framework, different types of communication were defined for consumers. In the ABM, each agent established connections with other household members and agents associated with the same node. The distribution network structure was then divided into clusters based on distance using the clustering algorithm proposed by Hajebi et al. (2013). Within each cluster, agents randomly communicated with several agents inside and outside their cluster (Fig. 2).

The theory of social impact (Latané 1981) has the ability to model the effects of social communication and external factors on consumers' water conservation attitudes. This theory was further developed by Latané and Nowak (1997) based on binary opinion dynamics and was applied in the ABM by Koutiva and Makropoulos (2016). In this study, social impact theory was employed as implemented by Koutiva and Makropoulos (2016), assuming the absence of demand management policies. The modeling of the effects of social relationships on NAR is presented in Appendix A. This appendix details the relationships for calculating the effects of social impact, the probability of opinion change, as well as the process of updating NAR. In this study, it was assumed that NAR affects the desirable demand range, which is explained in the following section.

2.3.1.3. Water demand components. In the conventional method for hydraulic analysis of WDNs, an up-bottom approach is employed whereby nodal demand is assigned based on per capita averages. This approach typically overlooks spatial and behavioral variability among nodes arising from differences in population density, demographic composition, consumer categories, and individual water use behaviors. Furthermore, nodal demand patterns are generally derived from aggregated demand patterns of the entire network (Fiorillo et al., 2020), regardless of the water demand behavior of consumers at each node. In the proposed framework, since the water use components were determined based on demand components, the demand components should be incorporated directly into the hydraulic analysis of WDNs. Water end-use models (Buchberger and Wu, 1995; Alvisi et al., 2003; Blokker et al., 2010; Creaco et al., 2016), by a bottom-up approach, capture the diversity of water demand components based on water use behavior and have the ability to be integrated by other models such as the ABM (Liu et al., 2016). Therefore, in the water demand simulator sub-model, the Water end-use model developed by Blokker (2010) was utilized. This algorithm generates demand components solely based on statistical information and individual characteristics, without the need for flow measurement prediction (Blokker, 2010; Blokker et al., 2010, 2017).

In the algorithm developed by Blokker (2010), the total daily water demand of a household, Q , is calculated using Eq. (2).

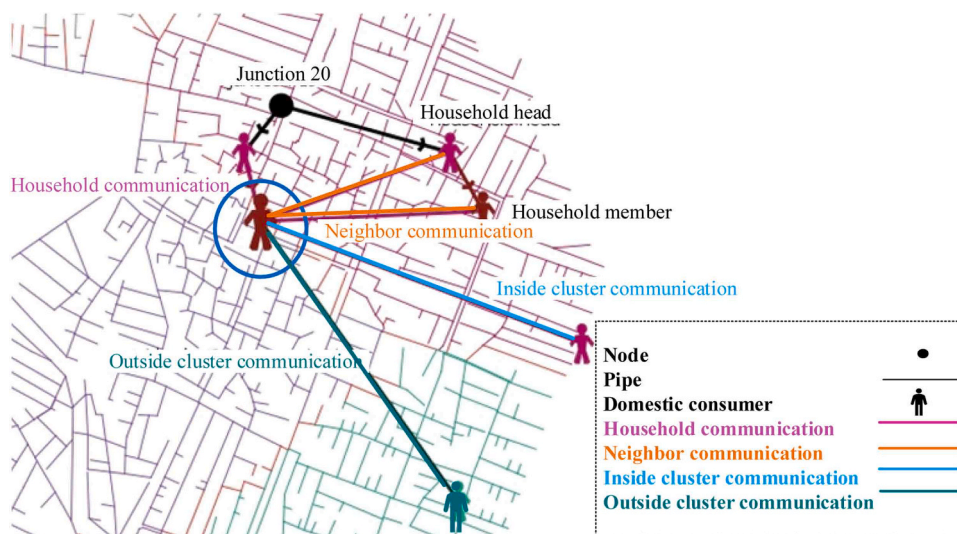


Fig. 2. Consumers' social communication in a WDN.

$$Q = \sum_{k=1}^M \sum_{j=1}^N \sum_{i=1}^{F_{jk}} B(R_{ijk}, D_{ijk}, \tau_{ijk}) \quad (2)$$

The block function $B(R_{ijk}, D_{ijk}, \tau_{ijk})$ is defined using Eq. (3):

$$B(R_{ijk}, D_{ijk}, \tau_{ijk}) = \begin{cases} R_{ijk} & \tau_{ijk} \leq t < \tau_{ijk} + D_{ijk} \\ 0 & \text{Elsewhere} \end{cases} \quad (3)$$

where Q denotes the total daily water demand of a household, k is the index for water use activities ranging from 1 to M , j is the index for household members ranging from 1 to N , i is the index for the occurrence of each water use activity ranging from 1 to F_{jk} (the number of repetitions of activity k for consumer j), R_{ijk} the demand intensity flow rate for consumer j at the i -th repetition of activity k , D_{ijk} refers to the demand duration for consumer j during the i -th repetition of activity k , and τ_{ijk} refers to the start time of demand for consumer j at the i -th repetition of activity k . $B(R, D, \tau)$ is a block function that equals R from time τ to $\tau + D$, and zero otherwise. The household demand pattern over a day is obtained from the sum of B for all N family members, M water use activities, and F repetitions of each activity.

In this study, the water use activities were divided into two main categories, including individual and household activities. Individual activities encompassed showering, toilet tap, and bathroom tap activities (washing and shaving, brushing teeth), which are non-volumetric activities. Household water use activities were divided into three sub-categories, which were: (1) non-volumetric activities, including kitchen tap activities (doing dishes, washing hands, and others) and outside tap activities (garden and others), (2) volumetric person-based activities, consisting of one kitchen tap activity (consumption), and (3) uncontrolled volumetric activities, including the use of the washing machine. For each individual and household activity, the frequency (F), intensity (R) (liters per second), and duration (D) (seconds) were determined based on the respective probability distribution functions provided in the study by [Blokkeer \(2010\)](#). In addition, the time of water demand (τ) (minute of the day and night) was assigned based on the consumer diurnal pattern, following the approach employed in the study by [Blokkeer \(2010\)](#).

2.3.1.4. Nodal water demand and water use pattern. In this study, nodal water use patterns and nodal demand patterns were obtained by a similar process through aggregating water demand or use components of consumers associated with each node. For the demand pattern, the process was as follows:

Since the hydraulics of the WDN was analyzed on a minute scale, a time series of 1440 min was defined for the nodal demand patterns. The t -th term of the time series was equal to the sum of the demand of nodal consumers in the t -th minute of the day and night. Since the flow intensity specified by the water end-use model was in terms of liters per second, and the nodal demand pattern was on a minute scale, it was necessary to calculate the equivalent flow intensity in terms of liters per minute. At first, the equivalent flow intensity was obtained for each activity, and then was added to those sentences of the time series that correspond to the times of doing the activity. By repeating this process for all nodal consumer activities, the nodal demand pattern was obtained.

Then, these nodal patterns were exchanged between the ABM and the hydraulic simulation sub-model, by an extended [Hajebi et al. \(2013\)](#) algorithm to execute an extended period of DDA on a minute time scale.

2.3.1.5. Desirable demand range. Consumers expect the outflow intensity of the tap to match their desirable demand. Unlike other studies that considered a fixed unit intensity flow for the desirable demand, this study defined a range of intensity values for the desirable demand of each water use activity. In other words, the flexibility of consumer behavior in the face of the state of the WDN service was modeled based on the desirable demand range. If the outflow intensity of the tap was not met by the desirable demand

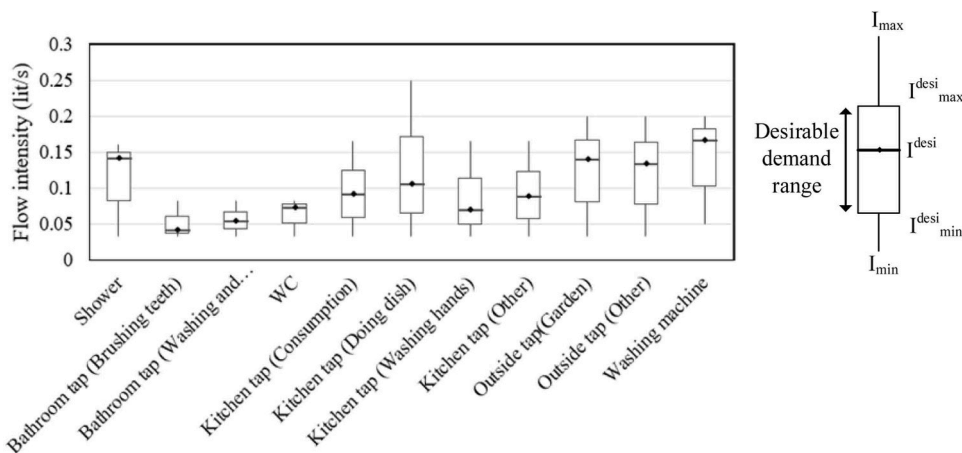


Fig. 3. An example of the consumer desirable demand range in one repetition of each water use activity.

range, the consumer adjusted the tap opening to align the outflow intensity within the range. The flow intensity, obtained from the water end-use model, was considered as desirable demand intensity (I^{desi}). The desirable demand range was defined asymmetrically around the desirable demand intensity based on the level of water conservation attitude. For example, the range is illustrated in Fig. 3 for a consumer with an NAR equal to 0.55. The upper and lower bounds of the desirable demand range, are referred to as maximum desirable demand intensity (I_{max}^{desi}) and minimum desirable demand intensity (I_{min}^{desi}), respectively.

Consumers for each water-use activity calculated these bounds based on their NAR, according to Eq. (4) and Eq. (5), as follows:

$$I_{min}^{desi} = I_{min} + (1 - NAR)(I^{desi} - I_{min}) \quad (4)$$

$$I_{max}^{desi} = I_{max} - NAR(I_{max} - I^{desi}) \quad (5)$$

here NAR is the normalized attitude rank, which was explained in the *consumers and characteristics* section, and two parameters, I_{min} and I_{max} , represent the minimum and maximum unconditional desirable demand intensities. The minimum (I_{min}) and maximum (I_{max}) unconditional desirable demand intensity was defined as the flow intensities that the entire community of consumers believed that it is not possible to perform the activity by intensities lower and higher than them, respectively. In Table 1, values for these parameters are presented for eleven water-use activities. The values of the minimum unconditional desirable demand intensity (I_{min}), desirable demand intensity (I^{desi}), and maximum unconditional desirable demand intensity (I_{max}) were extracted from the study by Blokter (2010) and the report published by Isfahan Water and Wastewater Company (2016).

In Eq. (4) and Eq. (5), the desirable demand range during each repetition of a water use activity is defined based on the consumer's conservation attitude, represented by the NAR parameter. A higher NAR indicates a stronger conservation intention, expanding the desirable demand range toward lower flow rates. In other words, consumers with higher NAR values are more willing to reduce their water use under low-pressure or shortage conditions without experiencing dissatisfaction, and their consumption is less affected by excess pressure. Conversely, lower NAR values correspond to a desirable demand range that extends more toward flow rates above the desirable demand intensity. In such cases, consumers tend to increase their water use to a greater extent under excess pressure and reduce it reluctantly under low-pressure conditions. The relationship between the desirable demand range and consumer behavior is further discussed in the *Water Use Behavior Algorithms* subsection.

2.3.2. Water use behavior simulator sub-model

For each activity, it was assumed that consumers adjusted the tap opening out of their habit and then were incentivized to adjust the tap opening based on their water conservation attitude. Consumers with high water conservation attitude had less motivation to increase the tap opening in low-pressure conditions, also they had more incentive to reduce the tap opening in the presence of excess pressure. On the other hand, consumers who were not willing to conserve water, had more motivation to increase the opening tap in low-pressure conditions, and had less incentive to reduce the opening tap in the presence of excess pressure. In general, consumers with more water conservation attitude showed better flexibility and adaptability in facing the WDN service conditions. In this study, the water conservation intention of consumers was considered dependent on their water conservation attitude. In this study, the effect of water conservation attitude on water use was modeled indirectly, by defining the desirable demand range according to the water conservation attitude. Therefore, the water use behavior of consumers was modeled based on the desirable percentage of the opening tap, the desirable demand range, and the state of the WDN service. The process of determining the desirable percentage of the opening tap, and the state of the WDN service are explained below, and after that, the water use behavior algorithms for determining water use components are presented.

2.3.2.1. Desirable percentage of the opening tap. In each water use activity, consumers open the tap out of their habit as much as the desirable percentage of the opening tap (O^{desi}). It was assumed that the desirable percentage of the opening tap represents the amount

Table 1

Desirable demand intensity and minimum/ maximum unconditional desirable demand intensities for water end-uses.

End-use type	Subtype	Minimum unconditional desirable demand intensity (lit/s)	Desirable demand intensity (lit/s)	Maximum unconditional desirable demand intensity (lit/s)
Bathroom tap	Washing and shaving	0.033	0.042	0.08
Kitchen tap	Brushing teeth	0.033	0.042	0.083
	Consumption	0.033	0.083	0.166
	Doing dishes	0.033	0.125	0.25
	Washing Hands	0.033	0.083	0.166
	Other	0.033	0.083	0.166
Outside tap	Garden	0.033	0.1	0.2
	Other	0.033	0.1	0.2
Shower	-	0.033	0.123	0.16
Washing machine	-	0.05	0.167	0.2
WC	-	0.033	0.042	0.083

of opening at which the outflow intensity is equal to the desirable demand intensity (I^{desi}) under normal operating conditions. To determine the desirable percentage of tap opening, it was assumed that under normal operating conditions of the WDN and sufficient pressure, the maximum unconditional desirable demand intensity (I_{max}) was achieved at full tap opening, while the minimum unconditional desirable demand intensity (I_{min}) was reached at one-third tap opening (33 % opening). Based on these assumptions, the desirable percentage of the opening tap (O^{desi}) was determined according to the desirable demand intensity (I^{desi}), within the range of one-third tap opening (33 % opening) to full tap opening.

2.3.2.2. State of the WDN service. An important factor influencing consumers' water use behavior is the service level of the WDN, which depends on the supplied pressure. Therefore, to assess the state of WDN service experienced by each consumer, nodal pressure patterns determined by the WDN hydraulic simulator had to be shared with the corresponding consumers in the ABM. The algorithm developed by Hajebi et al. (2013), originally designed to import only the WDN structure into the NetLogo environment, was extended in this study to also import nodal pressure patterns into the ABM. In the traditional methods for WDN hydraulic analysis, the state of the WDN service is determined solely based on nodal pressure. In this aggregated approach, the state of WDN service is defined at the nodal level, while the spatial distribution of taps around the node is neglected. However, in this study, the state of the WDN service was defined at the tap level based on nodal pressure, by incorporating the vertical and horizontal distribution of taps around each node. This approach enables more detailed considerations, yielding a more realistic representation of the WDN service state (Schück and Lansley, 2018; Díaz et al., 2024).

This study assumed that in each repetition of water use activities, the relationship between tap outflow intensity and the pressure head behind the water meter was definable through the orifice plate equation (ISO, 2003). This equation was modified to account for energy loss and the desirable percentage of tap opening. The relationship between flow area and tap opening percentage varies depending on the tap type. However, regardless of tap type, the cross-sectional flow area for a partially opened tap was assumed to be $O^{desi} \times A_2$, where A_2 represents the outlet tap area. Therefore, the available water flow intensity (I^{avl}) for the desirable percentage of tap opening (O^{desi}) was obtained as Eq. (6) in terms of liters per second:

$$I^{avl} = \left(\frac{1}{\sqrt{\left(\frac{1}{O^{desi}^2} + 1.5 \frac{d_2}{d_1} \right) - \left(\frac{d_2}{d_1} \right)^4}} \right) A_2 \sqrt{2g(H_j^{ave} - h_v)} \quad (6)$$

where H_j^{ave} , represented the pressure head behind the water meter, which was considered to be equal to the average value of the nodal available pressure head (H_j) during the water use activity period (D), h_v was equal to the height difference between the tap outlet and the water meter, g was the gravitational acceleration constant, and O^{desi} was the desirable percentage of the opening tap. The pipe friction factor (f), along the flow path from behind the water meter to the tap outlet, was assumed to be 0.032 (Moody, 1944; Churchill, 1977). The equivalent pipe length for evaluating all minor head losses was assumed to be half the flow path length ($0.5 L$). Due to the lack of data on household building plumbing systems, the flow path length (L) was assumed to be proportional to the building area, which was considered a proxy for its horizontal dimension. Assuming water fixtures are distributed around the perimeter, the path length was approximated as the diagonal of a square with an area equivalent to that of the building. d_1 represented the service pipe diameter, where the diameters for the internal building pipes and tap pipes were assumed to be the same and equal to d_2 , based on field observations in the study area and the report published by the Isfahan Water and Wastewater Company (2016). In this equation, pipe diameters, pressure head, elevation, and length were given in meters, and time was given in seconds. By Eq. (6), consumers calculated the available flow intensity based on nodal available pressure for each repetition of the water-use activity to become aware of the state of the WDN service.

2.3.2.3. Water use behavior algorithms. Water use behavior algorithms are a section of ABM and determine the water use components, including flow intensity, duration, and start time. These algorithms were different for each water-use category that was explained in the *Water demand components* section. In the water use behavior algorithms, the consumers became aware of the WDN service status by comparing the desirable demand range with the available flow intensity, and then the water use components were determined based on consumer behavior. These behavioral responses were derived from field investigations and observations of residents' actions in the study area, reflecting the real-world strategies consumers use to adapt to water availability constraints. In ABM, these consumer behaviors are modeled using a set of if-then commands as follows:

- If the available flow intensity met the desirable demand range, water use activity was done in normal conditions by intensity equal to the available flow intensity.
- If the available flow intensity exceeded the desirable demand range, by decreasing the opening tap, the available flow intensity was reduced and met the desirable demand range.
- If the available flow intensity did not reach the desirable demand range, by increasing the tap opening, the available flow intensity was increased and met the desirable demand range.

- If pressure was not sufficient, the maximum available flow intensity did not reach the desirable demand range and sometimes did not even reach the minimum unconditional desirable demand intensity. In such cases, the consumer did activities by collecting water at that time, if possible. Otherwise, consumers adjourned the collection of water until the pressure was suitable.
- If the suitable pressure was not provided at any time, the consumer canceled the activity, and the water use flow intensity was set to zero.

Fig. 4 illustrates the consumers' actions with a specific desirable demand intensity (I^{desi}) and different NAR values, facing varying states of the WDN service, briefly. In this figure, the green area represents the consumer's desirable demand range, the orange area indicates conditions where the available flow intensity exceeds this range, and the blue, yellow, and red areas correspond to conditions where the available flow intensity falls below the desirable demand range, leading to consumer dissatisfaction with the hydraulic performance of the WDN.

Furthermore, in water use behavior algorithms, the dissatisfaction level of consumers due to inadequate network services was modeled. A negative point was assigned based on the type of disruption experienced by the consumer, including canceling the activity, performing the activity manually instead of using a machine, collecting water at another time, collecting water at the same time, performing the activity with undesirable intensity, and performing the activity after increasing the tap opening. Negative points were assigned to these behaviors in a sequential order from 1 to 6, where the most difficult behavior was given a score of 6 and the least difficult a score of 1. The sum of consumers' negative points indicates the consumers' dissatisfaction level.

3. Framework implementation on case study

The WDN of Gaz City has been chosen as the case study. Gaz is a historical city situated on a relatively flat plain and lacks natural features. The Gaz City is part of Shahin Shahr and Meymeh counties in Isfahan province, Iran (Fig. 5-a). The water supply in the WDN of GAZ City is gravity-fed from a reservoir at an elevation of 1610 m. Based on collected information, some characteristics of the WDN and the community in GAZ City are presented in Table S.2.

The network structure of the WDN of Gaz City was imported into the NetLogo environment by utilizing the calibrated hydraulic simulation model of the WDN. In the case study, due to a lack of accurate information, the location of the consumers is not known, thus the number of consumers in each node was determined by dividing the nodal base demand by per capita water use. Consumers' properties were assigned according to the statistical information obtained from the census, except for cognitive beliefs, which were weighted using a uniform probability distribution function.

The structure of the WDN of GAZ was divided into four clusters (Fig. 5-b), and consumers communicated with each other inside and outside the clusters. Then, the consumer's water conservation attitude was assigned based on their properties and communications. Some characteristics of the modeled community are presented in Table S.3. The developed framework analyzed the hydraulics of the WDN. The results of applying the proposed framework to the WDN of the Gaz City and consumer society are presented in the following sections.

4. Results and discussion

4.1. Results of the water demand simulator sub-model

The first step in the water demand simulator sub-model involves determining consumers' water demand components based on the probability distribution functions of the frequency, intensity, duration, and start time. Two consumers were selected as examples from the consumer community, consumers #8981 and #7489. They have identical total demand and different water conservation attitudes. Their characteristics, along with the results obtained from various sections of the proposed framework, will be presented in the

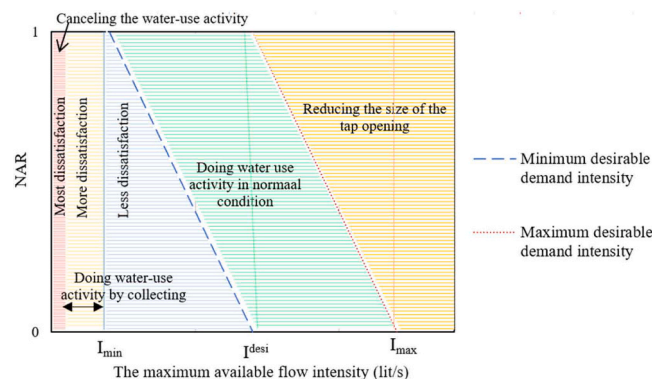


Fig. 4. Determining consumers' actions with different NAR facing different states of the WDN service.

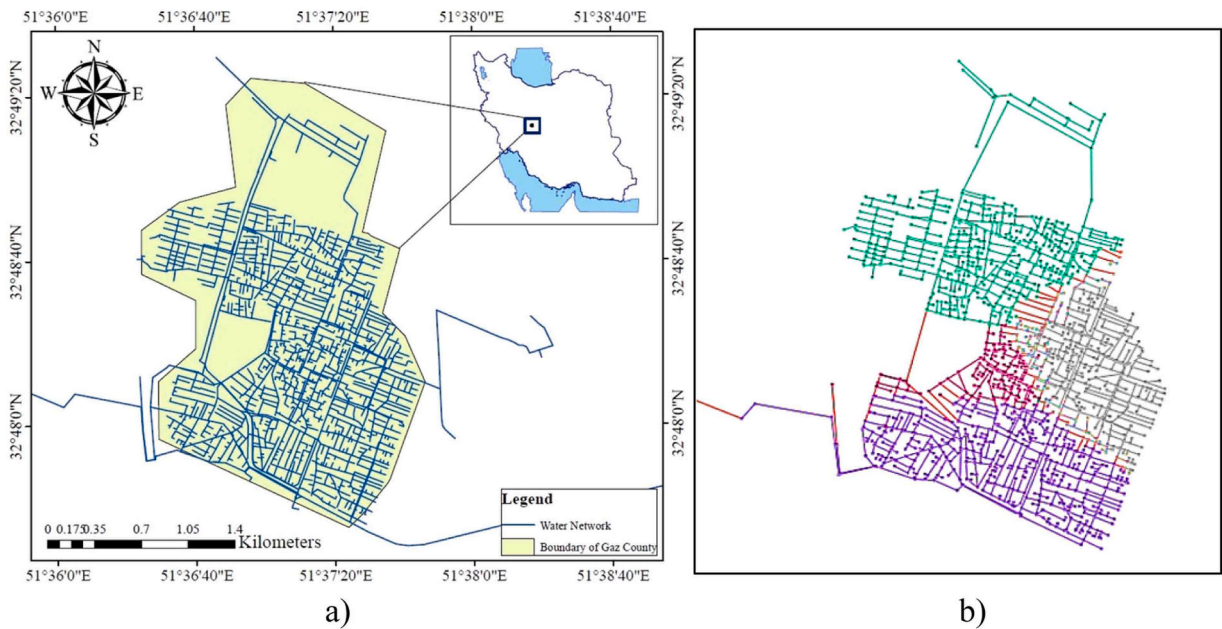


Fig. 5. a. The WDN of GAZ City, b. The WDN clusters.

following sections.

Fig. 6 presents the results obtained by the water demand simulator sub-model for these consumers. This figure compares the desirable water demand volumes of these two consumers for various water use activities. Although the total demand was about 263 liters for both consumers, their demand components, including the frequency, intensity, duration, and start time, differed in the scale of water use activities. As shown in this figure, consumer #8981 has a higher water demand than consumer #7489 for activities involving the outdoor tap (garden), kitchen tap (other), and shower, both consumers exhibit similar demand for the washing machine, whereas for all other activities, the demand of consumer #8981 is lower than that of consumer #7489. In addition to the difference in demand for various activities, most of the water used was related to showers and doing dishes for both consumers.

In the second step, for each repetition of water use activities, the desirable demand range was determined using Eq. (4) and Eq. (5), based on the desirable demand intensity and consumers' normalized attitude rank (NAR). Fig. 7 illustrates the desirable demand range for one repetition of various water use activities for both consumers to demonstrate the capability of the model to simulate the diversity of the demand behavior of consumers in the water use activity scale. The water conservation intention of consumer #8981 (NAR = 0.55) is higher than that of consumer #7489 (NAR = 0.45). In activities where both consumers exhibit similar desirable demand, their desirable demand ranges can be compared. In the dishwashing activity, the desirable demand intensity is 0.106 liters per second for both consumers. The upper bound of the desirable demand range for consumer #7489, due to the lower conservation attitude, is more stretched toward maximum desirable unconditional demand intensity ($I_{\max}$) compared to consumer #8981. Conversely, the lower bound of the desirable demand range for consumer #8981, due to the higher conservation attitude, is more stretched toward minimum unconditional desirable demand intensity ($I_{\min}$).

In the third step, the nodal desirable demand range was determined by aggregating the lower and upper bounds of the desirable

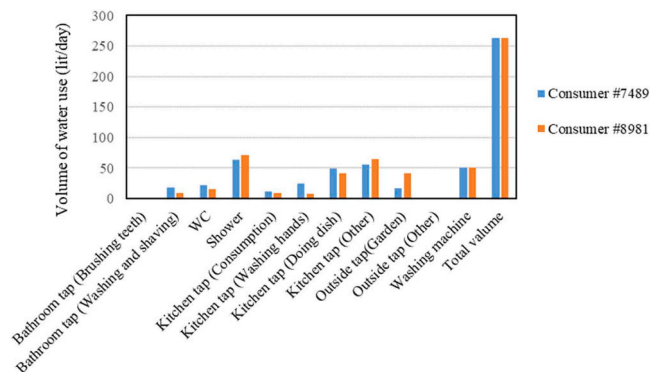


Fig. 6. Water demand volume of consumers #7489 and #8981 in various water use activities along with their total water demand volume.

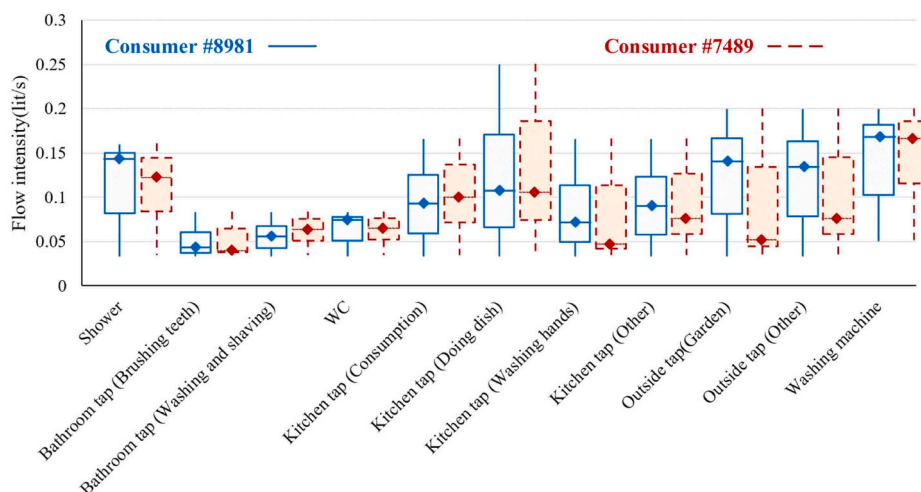


Fig. 7. Desirable demand range for one repetition of water use activities for consumers #7489 and #8981.

demand range related to consumers' water use activities associated with that node. Desirable demand intensities are shown on the desirable demand range for consumer #8981 in Fig. 8, through day and night.

Fig. 8 also displays the nominal demand pattern, which is the demand pattern traditionally used in the hydraulic analysis of the study area's WDN. In the nominal demand pattern, the daily variation in nodal demand is typically defined cumulatively for all nodal consumers by applying different coefficients to the base nodal demand at various time steps. As observed, the nominal demand pattern values are very small. This is because, in the nominal pattern, the daily demand is continuously distributed throughout the day. In contrast, in the desirable demand pattern, similar to the actual demand pattern of the consumer, demand remains zero for most of the time and becomes non-zero only during periods of activity. Additionally, this results in more pronounced demand variations in the proposed model than in the nominal demand pattern, and the peak demand values are modeled higher than those in the nominal pattern.

4.2. Results of the water use behavior simulator sub-model

Water use components in each repetition of activities were determined in the water use consumer behavior simulator sub-model. Table 2 lists the supply pressure and water use obtained from the proposed model for six consumers at reservoir heads of 1625 m and 1635 m. These results demonstrate the ability of the model to simulate the effect of the diversity of consumer behavior and WDN service conditions on the water use of consumers. The performance of the proposed model was evaluated by comparing its results with those obtained from PDA. The demand values listed in the fifth column were obtained from the demand simulator sub-model. Average values of nodal pressure and total water use during the day and night were obtained from hydraulic analysis, presented in the sixth and seventh columns, respectively. The eighth and ninth columns also list the pressure and water use values resulting from PDA. In Simulation Case 1, the status of consumers #25871 and #20487, who have the same water conservation attitude, were compared. While the total demand of the two consumers was equal, they were different regarding water demand components. Both the proposed model and PDA estimated the water use of consumer #20487 more than consumer #25871 due to the difference in their nodal available pressure. This result demonstrates that the proposed model, similar to PDA, accurately simulates the effect of pressure, thus validating the proposed model. According to the Table 2, the proposed model estimates the water use difference between the two

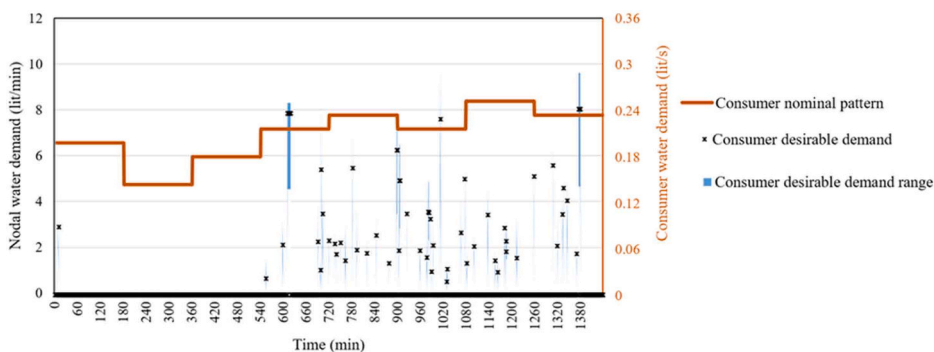


Fig. 8. Desirable demand intensity on desirable demand range along with nominal demand patterns for consumer #8981.

Table 2

Comparison of the results of the proposed model and PDA.

Sim. Cases	Res. head (m)	Cons.	NAR	Dem. (lit)	Ave. pres. (Model) (m)	Use (Model) (lit)	Ave. pres. (PDA) (m)	Use (PDA) (lit)
Sim. Case 1	1625	25871	0.5	202	28.9	175	29.22	184
	1625	20487	0.5	202	29.2	185	29.59	189
Sim. Case 2	1625	8981	0.55	263	32.19	277.8	32.66	258.1
	1625	7489	0.45	263	32.19	285.5	32.43	257
Sim. Case 3	1625	22928	0.55	215	28.24	224.6	32.33	213
	1625	24238	0.43	215	26.57	219.0	31.03	203
Sim. Case 4	1635	22928	0.55	215	38.23	247	41.7	215
	1635	24238	0.43	215	36.5	267.6	40.25	215

Sim. Case = Simulation Case, Res. head = Reservoir head, Cons. = Consumer, Dem. = Demand, Ave. pres. = Average pressure

consumers to be nearly twice that of the PDA. This result reveals that the proposed model more effectively captures the diversity in water use among consumers compared to PDA, as it accounts for the consumer's water demand based on the type of water use activity.

In Simulation Case 2, two consumers, #7489 and #8981, were selected. These two consumers, had identical total demand and supply pressure but differed in their NAR values (0.45 for #7489 and 0.55 for #8981). This distinction enabled a controlled comparison, isolating the impact of water conservation attitudes on water use and evaluating the differences between the proposed model and the PDA model. Their water demands were compared in the previous section (Figs. 6 and 7). As determined, their total demands were the same, while they were different in terms of water demand components. In PDA, only the pressure effect was observed, and the water use of consumer #8918 was estimated to be about 1 liter more than consumer #7489, notwithstanding behavioral differences and the type of water use activities. This low difference was due to the difference in their water use time pattern during the day and night. However, in the proposed model, consumer #8981 consumed almost 7.7 liters less water than consumer #7489 due to a greater tendency towards water use conservation. The observed differences in water use highlight the significance of NAR in shaping water use, illustrating how variations in consumer water conservation attitudes influence water withdrawal behavior. Although the simulated difference in water use between the PDA and the proposed model is small in Simulation Cases 1 and 2, it is important to note that this difference reflects only two consumers and could become significant when scaled to the population of approximately 24,000 in Gaz City. Therefore, the overall network performance is evaluated in the "Results of Different Reservoir Heads Scenarios" section, using both the proposed model and PDA.

In Simulation Cases 3 and 4, the water use of two consumers, #22928 and #24238, who demonstrated distinct water conservation attitudes and were subjected to different pressure conditions, was compared. In Simulation Case 3, at a reservoir head of 1625 m, both consumers' water use exceeded their respective demands. Consumer #22928, despite possessing a higher conservation attitude, consumed more water than consumer #24238 due to experiencing higher pressure. Similarly, under PDA, consumer #22928 used more water than consumer #24238 as a result of the greater pressure experienced. Notably, although the nodal pressures calculated by PDA were higher than those in the proposed model, both consumers' water use remained below their respective demands.

In Simulation Case 4, the same consumers were evaluated at a reservoir head of 1635 m, corresponding to higher pressure conditions than those in Simulation Case 3. Both consumers again exhibited water use exceeding their demand. Consumer #22928, who had a higher water conservation attitude, consumed less water than consumer #24238, despite experiencing higher pressure, whereas consumer #24238 had a lower conservation attitude and experienced lower pressure. These results indicate that the influence of water conservation attitudes becomes more pronounced under higher pressure conditions. In contrast, under PDA, although the nodal pressures were calculated to be higher than in the proposed model, the water use of both consumers matched their respective demands, regardless of differences in pressure or conservation attitudes.

4.3. Results of different reservoir heads scenarios

To evaluate the impact of operating conditions in Gaz City's WDN on consumer water use, the WDN's hydraulic performance was simulated under different reservoir heads, and consumer water use behaviors were analyzed under the influence of the different states of the WDN service. Table 3 lists the hydraulic analysis results for 1625–1645 m reservoir heads. The demand satisfaction ratio in the

Table 3

The results of the proposed framework for different reservoir heads.

Reservoir head (m)	Total consumer's negative points	Demand satisfaction ratio	Canceling the water-use activity	Collecting water at the same moment	Collecting water at another time
1625	251972	1.013	29245	10198	843
1630	6576	1.117	639	433	51
1635	0	1.129	0	0	0
1640	-	1.131	0	0	0
1645	-	1.130	0	0	0

entire WDN for reservoir head of 1625 meters was calculated to be 1.013, which was estimated based on demand and water use values of 3290,902 and 3333,683 liters per day, respectively. Although the calculated demand satisfaction ratio suggested that consumers' total water use in the entire WDN was almost equal to the total consumer demand, the total consumer negative point in this reservoir head was reported to be about 250,000. This high negative point indicated the inappropriate state of the WDN service and consumer dissatisfaction. The high consumer dissatisfaction, coupled with the demand satisfaction ratio greater than 1, indicated that the state of the WDN service was suitable for some consumers and unsuitable for others.

In the 1625 m reservoir head, although the demand satisfaction ratio of the WDN is more than 1, the demand satisfaction ratio was reported to be 0.45 for a three-person household with a demand of 379 liters. They did not receive a suitable WDN service to perform 22 activities. Fig. 9 compares the water use values obtained from the proposed model vs. demand. This figure shows that their water use is much lower than the demand during peak water use times. In GAZ City's WDN, the peak water use periods occurred between 10:00–14:00 and 16:00–20:00. As observed in Fig. 9, during off-peak periods such as 1320–1380 min, excess pressure resulted in higher water use compared to the demand.

Table 3 presents the social results of the proposed model for the behavior and actions of consumers facing the WDN service. As shown in the fourth column, at the 1625 m reservoir head, all consumers canceled about 29,000 of their water use activities, about 10,000 activities were performed by water collection at the same time, and about 800 activities were done by water collection at other times. At the 1635 m reservoir head, the number of activities that consumers could not perform as usual reached zero. The 1635 m reservoir head results indicate that the demand satisfaction ratio was 1.13; in other words, water use in the WDN was about 13 % more than demand. Also, the total negative point of consumers was 0; thus, the state of the WDN service in this reservoir head was favorable for all consumers during the day and night.

Table 3 shows the comparison results of the demand satisfaction ratio provided for different reservoir heads. As indicated, water use in the WDN negatively increased with every 5 m increase in the reservoir head. The demand satisfaction ratio corresponding to the 1625–1635 m reservoir head values implies that with increasing the reservoir head, consumer water use, and subsequently, the demand satisfaction ratio increased up to 24.6 %. On the other hand, in reservoir heads higher than 1635 meters, with increasing the reservoir head, no significant changes were observed in consumer water use and subsequently in demand satisfaction ratio, which was attributed to the control of the opening tap by the consumers in the conditions of high excess pressure.

Moreover, Fig. 10 depicts the WDN water use obtained from the proposed model for different reservoir heads in comparison to water demand. Here, each point corresponds to WDN water use and demand at a specific moment of the day. The orange (I_{min}^{desi}) and grey (I_{max}^{desi}) points representing the minimum and maximum desirable demand intensity graphs, are estimated by aggregating the lower and upper bounds of consumers' desirable demand range. If data points fall below the orange graph (I_{min}^{desi}), the state of the WDN service is highly inappropriate. This figure shows that all points corresponding to the 1635 m reservoir head are either on or above the water use–water demand equality line. However, a significant number of points associated with the 1625 m reservoir head fall below the minimum desirable demand graph, during peak demand periods. This indicates that during this period, the WDN hydraulic performance was weak, and meeting all consumers' demands was not feasible.

In Table 3, it is reported that at the 1635 m reservoir head, consumers experienced no dissatisfaction with the network's service. In Fig. 10, the consistent results for both 1625 m and 1635 m reservoir heads during off-peak times show that consumers at the 1625 m reservoir head also had no dissatisfaction during off-peak periods. As a result, dissatisfaction at the 1625 m reservoir head occurred only during peak demand periods, due to service disruptions.

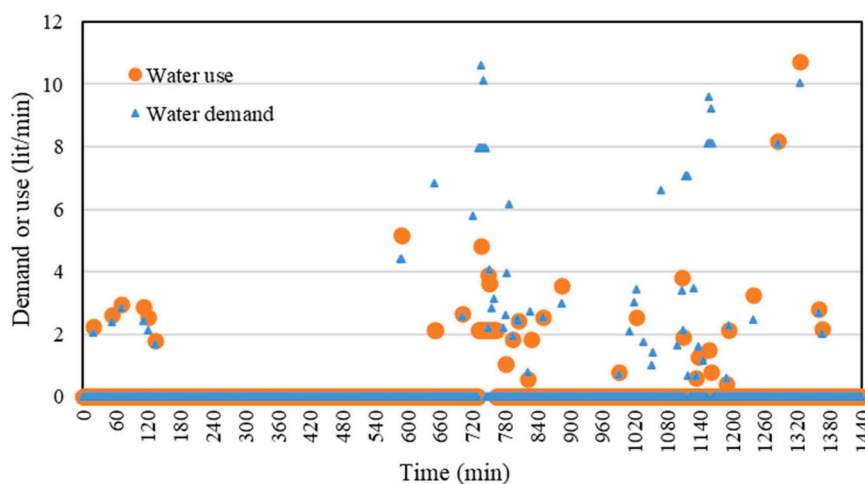


Fig. 9. Comparison of water demand and water use for a three-person household.

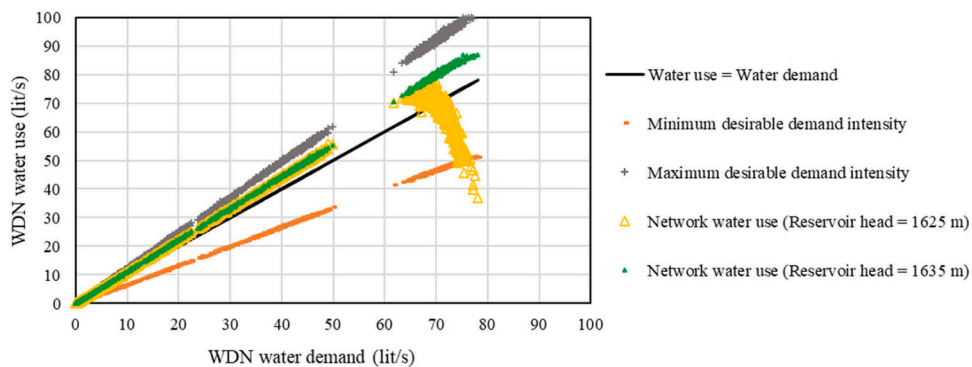


Fig. 10. The water use versus demand of WDN at different reservoir heads.

4.4. Results of different water conservation attitude scenarios

The impact of changing water conservation attitudes on consumer water use and the hydraulics of the WDN were examined by defining three scenarios (i.e., A, B, and C) under the 1635 m reservoir head. As presented in Table 3, the operational conditions of the WDN are normal at this reservoir head. The consumers had different desires and heterogeneous attitudes toward water use conservation. In Scenarios A, B, and C, the water use conservation attitude of consumers was homogeneously set equal to the minimum, average, and maximum values, respectively. In each of the three scenarios, the maximum and minimum demand were different according to the consumers' attitudes. For example, the desirable demand range at one repetition of various activities for one consumer, calculated using Eq. (4) and Eq. (5), is depicted in Fig. 11, for the extreme scenarios: Scenario A and Scenario C. This figure illustrates that in Scenario A, consumers do not consider a flow intensity lower than the desirable demand intensity as acceptable, whereas in Scenario C, they do not consider a flow intensity higher than the desirable demand intensity as acceptable. This discrepancy results in variations in water use outcomes and, subsequently, the hydraulic condition of the WDN.

Fig. 12 illustrates the daily variations in network water use obtained from the proposed model for each scenario. Additionally, the water use variations from PDA and the WDN's demand fluctuations are presented to enable a comparison with the proposed model's results. In Scenario A, consumers were reluctant to conserve water, and subsequently, water withdrawal was not less than demand unless there was not enough water. Therefore, all the consumers located in the nodes with overpressure withdrew more water than the demand due to their unwillingness to conserve water. It caused the WDN's service to become undesirable for some other consumers to the extent that they did not have the possibility of water withdrawal as much as the demand. Water use points on the graph linked to Scenario A in Fig. 12 show that, during certain peak times, the water use was lower than the demand. In this scenario, consumers with no desire to conserve water did not exhibit flexibility in the face of reduced service of the WDN and expressed dissatisfaction earlier. Thus, for this scenario, the negative point reported in the entire WDN was 637424. The results of this scenario indicate that a reduction in consumers' water conservation attitudes can shift the WDN's operational conditions from normal to abnormal at a reservoir head of 1635 meters.

In Scenario B, the water use conservation attitude was defined as moderate. In this scenario, the demand satisfaction ratio in the WDN was estimated to be greater than 1, and the negative point of the entire WDN was equal to 0, indicating consumer satisfaction

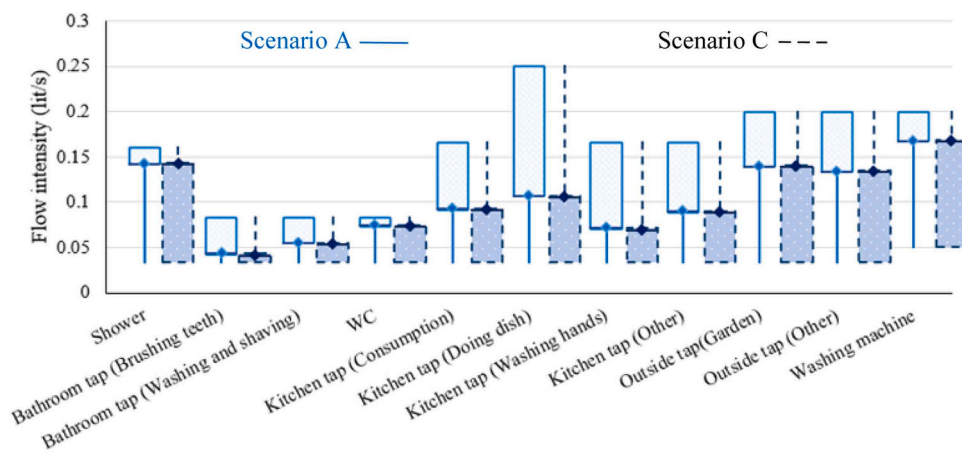


Fig. 11. Consumer's desirable demand range in one repetition of each water use activity for scenarios A and C.

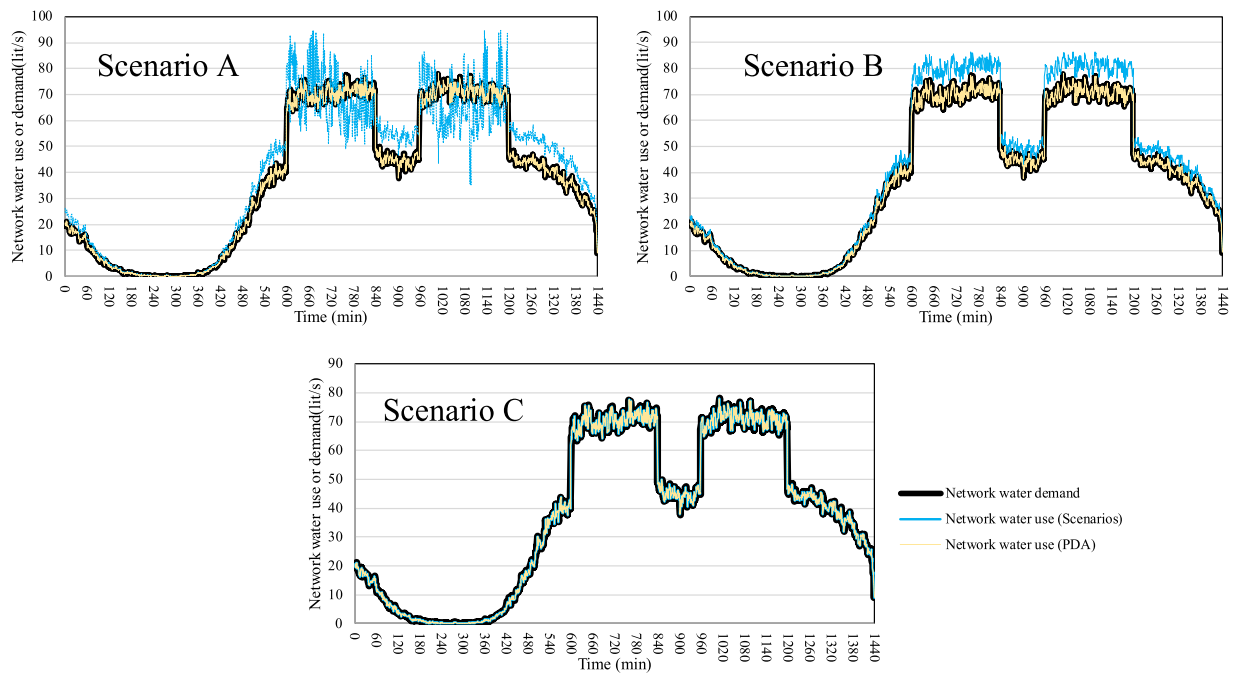


Fig. 12. Comparison of WDN water use variations from the proposed model with PDA results and demand throughout the day and night (Scenarios A, B, and C).

with the WDN service. Furthermore, the graph for Scenario B shows that the WDN's water use obtained from the model exceeds WDN's demand, approximately at all times, indicating a satisfactory service state for all consumers. Comparing the results of Scenario A and Scenario B shows that increasing consumers' conservation attitudes shifted the WDN's operating conditions from abnormal to normal.

The desirable demand range for Scenario C is shown in Fig. 11. The graph linked to Scenario C in Fig. 12 demonstrates that the WDN's water use aligns with the WDN's demand at all times. In this scenario, the demand satisfaction ratio was estimated to be 1, and the total negative points of consumers was 0, indicating water withdrawal did not take place more than the demand in addition to the satisfaction of the consumer community.

The results of different scenarios were changed by changing the reservoir head and subsequently changing the state of the WDN service. For the 1630 and 1635 m reservoir heads, shifting from Scenario B to Scenario C, and subsequently, increasing consumers' willingness for water use conservation, water use during the day and night decreased to 387 and 426 m³, respectively, constituting about 13 % of the total desirable demand of the WDN (3290,902 liters per day). The reduced water use volumes associated with the two reservoir head levels also indicate that the effect of water conservation attitudes on reducing water use has increased with increasing the reservoir head due to overpressure in the WDN.

In Fig. 12, a comparison of the results from the three scenarios with those from the PDA model reveals that, in the PDA model, the operating conditions remained normal at all times, with water use consistently matching demand, irrespective of consumers' attitudes. In the graph for Scenario C, the results from both the proposed model and the PDA model are identical. This can be attributed to the fact that, at a reservoir head of 1635 m, normal operating conditions were sustained, meaning none of the consumers displayed behaviors indicative of pressure-deficient conditions. Moreover, due to the consumers' high conservation attitudes, their water use was precisely aligned with their demand. These two factors are consistent with the simplifications in the PDA model, resulting in similar outcomes from both models. This serves as a validation of the proposed model.

5. Discussion

5.1. Insights from the model at the individual scale

In this study, consumer water use was simulated at the microscale, necessitating the modelling of demand at the same resolution (Giustolisi and Walski, 2012). To this end, the stochastic bottom-up approach developed by Blokker (2010) was employed, enabling explicit representation of heterogeneity in individual water demand. A comparison between two consumers, #8981 and #7489, showed that although the total daily demand volumes were identical, the water demand volumes across different water-use activities varied (Fig. 6). This variation was attributed to differences in their disaggregated demand profiles, specifically in the frequency, intensity, and duration of water-use events. This micro-scale simulation captured discontinuous demand patterns with pronounced peaks, in contrast to the smoothed and continuous profiles typically generated by conventional models (Fig. 8). In addition, the framework accounted for variability in consumer responses to service conditions. This behavioral heterogeneity was modelled through

individual water conservation attitudes. Based on each consumer's NAR, a desirable demand range was defined for each repetition of each water-use activity (Eq. 4 and Eq. 5), reflecting the consumer's flexibility in adapting to fluctuations in WDN pressure. In Fig. 7 and Fig. 11, it was illustrated that as NAR increases, consumers show greater flexibility in accepting lower flow rates as adequate and exhibit higher tolerance toward reduced water supply. In demand pattern, by combining each consumer's desirable demand intensity with their acceptable demand range, water use at each time step is no longer represented by a fixed value but rather by a flexible interval that reflects the consumer's capacity to adjust demand while maintaining satisfaction (Fig. 8). This concept was also extended to the WDN level (Fig. 10), enabling the assessment of total water use in the WDN relative to the aggregated desirable demand ranges. This representation offers deeper insight into how consumers adjust their water use during peak and off-peak periods in relation to their minimum desirable demand and maximum desirable demand.

Furthermore, WDN service conditions were modeled at the micro scale, capturing a more realistic consumer experience during every instance of water-use activities. For each repetition, the outflow rate was determined using Eq. (6), incorporating characteristics such as habitual tap opening, fixture elevation, and the residential floor level. This approach aligns with recent studies emphasizing that the relationship between water withdrawal and nodal pressure is significantly influenced by downstream secondary infrastructures, including building plumbing and service pipes (Schuck and Lansey, 2018; Burkhardt et al., 2023; Díaz et al., 2024). The influence of faucet opening levels on water supply has also been consistently emphasized (Ferreira et al., 2018).

Simulating the effects of faucet opening levels and fixture elevation at the scale of individual water-use activities enabled the representation of a wide range of consumer behaviors in response to varying service conditions, including drawing water from lower taps, storing water, or adjusting tap openings. Fig. 4 illustrates how these behavioral variations are derived by comparing the supplied flow rate with the consumer's desirable demand range. Such behavioral variability under low-pressure conditions has also been recognized in previous research. For example, Díaz et al. (2024) modeled consumer responses to reduced pressure, showing that individuals may adapt by using lower taps. Other studies have identified household water storage as a common coping strategy during intermittent supply (Giustolisi et al., 2014; Gonzalez et al., 2020).

In the present framework, defining the desirable demand range based on varying consumer attitudes enabled the simulation of heterogeneous consumer behavior under specific service conditions. In this context, a comparison between consumers #8981 and #7489 further demonstrated that, under identical pressure conditions, the consumer with a higher NAR exhibited more water-saving intention and consumed a total of 7.7 liters less (Table 2). The importance of attitude as a key determinant of water conservation behavior and intentions has been confirmed in previous studies (Si et al., 2022; Shahangian et al., 2024). Furthermore, several studies have supported that water-saving behaviors are directly associated with reductions in household water consumption (Fielding et al., 2012; Cook et al., 2018).

5.2. Insights from the model at the WDN scale

The proposed framework enables evaluation of WDN performance by integrating both hydraulic and social perspectives, addressing a limitation of indicators like the demand satisfaction ratio that focus solely on hydraulic aspects of service provision. The results in Table 3 show that, under certain operating conditions, even when this technical metric suggests adequate performance, the proposed model indicates that consumers experience inadequate service due to low pressure. This is evidenced by reported dissatisfaction and adaptive behaviors such as water storage and the cancellation of water-use activities. Household-level analysis supports this finding. For example, while the demand satisfaction ratio exceeded one, a three-person household met less than half of its daily water demand (Fig. 9). This discrepancy between demand satisfaction ratio and consumers' experience arises from variations in water use behavior: some consumers use more than their demand, while others fail to meet their minimum desirable demand due to insufficient pressure. In addition, the proposed framework offers valuable insights into how varying service conditions throughout the day influence water use. By comparing water demand and use at both household (Fig. 9) and WDN scales (Fig. 10 and Fig. 12), a detailed assessment of water use patterns during peak and off-peak hours was provided. The results indicated that during peak hours, water use fell below demand, potentially leading to consumer dissatisfaction. Conversely, during off-peak periods characterized by excessive pressure, water use exceeded demand, primarily due to limited incentives for conservation.

Moreover, the results of the proposed model highlight that the integration of consumer water conservation behaviors with WDN hydraulics offers a more comprehensive understanding of the dynamics within water distribution systems. Scenario analyses, as illustrated in Fig. 12, indicate that meaningful improvements in WDN performance can be achieved through behavioral changes alone, without the need for modifications to WDN operation. Specifically, at a reservoir elevation of 1635 m, the framework predicts that enhancing consumers' conservation attitudes can reduce total water use by approximately 13 %. Conversely, scenarios characterized by weak conservation attitudes show that excessive water use by some consumers adversely impacts service quality for others and increases overall dissatisfaction. It demonstrates that hydraulic performance is shaped not only by engineering parameters but also by individual consumer behavior in response to service conditions at the micro-scale of water-use activities. Consistent with these results, Liu et al. (2016) confirmed that conservation behaviors influence the hydraulics of both WDNs and sewer systems. Similarly, Kandiah et al. (2014) and Tourigny and Filion (2018) demonstrated that the adoption of reuse systems and water-saving technologies significantly impacts WDN hydraulic performance. These findings underscore the necessity of integrating social dynamics alongside technical parameters to enhance the operational efficiency of WDNs.

5.3. Adaptability of the model in WDN analysis

The proposed model was developed to simulate micro-scale consumer water use behavior and was implemented with a one-minute

time step for a medium-sized distribution network serving approximately 24,000 individuals. The model was designed to be adaptable to various network configurations. However, for application in larger systems, different consumer groups (such as industrial and public users) and their specific water demand patterns need to be defined and incorporated using an appropriate water end-use model, such as the [Blokker \(2010\)](#) model. Although the computational load may increase with a larger population, the implementation is not expected to encounter significant limitations, as long as sufficient computational resources are available.

The implementation of the proposed model in more complex WDNs involving pumps, valves, and intermittent supply requires further consideration. In such WDNs, two scenarios can be distinguished based on whether the operation of pumps and valves is preplanned or automatically controlled in response to variations in water use and pressure within the WDN:

1-In cases where the operation of control devices, such as pumps and valves, or the water supply schedule is predetermined and independent of consumer behavior, the proposed model is fully compatible. Under these conditions, since pump start-stop cycles, valve operations, or intermittent water supply occur at predetermined times, the model exclusively simulates the interaction between consumer water use and the hydraulic state of the WDN. Additionally, the hydraulic simulation sub-model, implemented using the WNTR package, is capable of performing hydraulic analysis of the WDN with the existing configuration.

2-In conditions where the operation of control devices such as pumps and valves is dynamically regulated in response to pressure or water use, a bidirectional dependency emerges between consumer behavior and network operation. Under these circumstances, in addition to the feedback loop between consumer behavior and the hydraulic state of the WDN, a secondary feedback loop develops between the control device schedule and consumer behavior. Furthermore, in intermittent water supply systems where scheduling is adapted in response to actual consumption, a similar secondary feedback loop is established between consumer water use and the water supply scheduling mechanism. The presence of this additional feedback loop increases the complexity of the simulation process and presents greater challenges for numerical convergence.

5.4. Applicability of the model for WDN management

The proposed framework is applicable in the management of WDNs. The present model enables the concurrent assessment of WDN performance and consumer satisfaction by simultaneously incorporating behavioral and technical factors in shaping water use. It supports water utilities in developing a more accurate understanding of how operational conditions influence consumer experiences, particularly under low-pressure scenarios that prompt responses such as water storage, withdrawal from lower taps, or the avoidance of consumption. The present framework enables a comprehensive evaluation of operational conditions from the consumer's perspective across both temporal and spatial dimensions by analyzing actual consumption patterns in relation to demand profiles, alongside consumers' responses to service quality. It thus provides a solid foundation for assessing equity in service delivery and supporting informed, fair, and effective decision-making.

In addition, the proposed model is applicable to the management of water scarcity through demand-side policies. These strategies are often developed and implemented using top-down approaches, with limited attention to consumer acceptance and cooperation. However, such hierarchical-oriented perspectives usually fail to anticipate the actual consequences of policy interventions. Therefore, A shift from conventional top-down implementation toward a bottom-up perspective is necessary to enable a more accurate and realistic evaluation of policy effectiveness. The application of the proposed framework, grounded in a bottom-up approach, facilitates the simulation of consumer satisfaction and behavioral responses to policy measures. This capability supports the assessment of the enforceability and practical viability of water management strategies under real-world conditions.

For instance, under conditions of water scarcity, one of the most commonly employed strategies is to reduce pipeline pressure and restrict water delivery within WDNs ([Lee and Schwab, 2005](#); [Olmstead and Stavins, 2009](#)). The proposed model can assist decision-makers in the effective implementation of such policies by providing a trade-off between the extent of pressure reduction and consumers' behavioral responses and dissatisfaction. Simultaneously, demand management policies aimed at promoting water conservation attitudes have gained increasing attention in the context of water scarcity. By integrating both technical and behavioral dimensions, the proposed framework enables the simultaneous assessment of the impacts of pressure reduction policy and the implementation of attitude-strengthening strategies. Under these conditions, the proposed framework facilitates understanding of how consumers' attitudes toward water conservation can enhance cooperation and encourage increased participation in water-saving behaviors.

5.5. Limitations and suggestions for future research

In this study, some simplifying assumptions were incorporated to calculate the desirable tap opening percentage, define the desirable demand range, and assess the state of WDN service. While these assumptions allowed the model to be implemented efficiently, further studies are encouraged to validate these assumptions and enhance modeling accuracy. For instance, parameters related to consumer habits could be introduced to better represent the desirable demand range and tap opening behavior, enabling more realistic simulations. Additionally, incorporating more accurate pipeline characteristics could improve the assessment of the WDN's service state. Conducting sensitivity analyses on key parameters in these contexts could provide deeper insights into the model's robustness and the relative importance of individual parameters. This approach is recommended for future studies to further enhance understanding and guide model improvements. A notable limitation of this study is the absence of behavioral theories and empirical psychological data. Future studies could integrate behavioral frameworks such as the Theory of Planned Behavior, and empirical evidence to enhance the conceptual foundation of the model and improve its representation of consumer water use intention and behaviors.

Applying the proposed framework to larger and more complex WDNs also represents a valuable avenue for future research. This includes integrating diverse consumer groups, such as industrial, commercial, and public users, along with their respective water demand profiles, using detailed end-use models. It is also recommended that the proposed model be applied to WDNs in which the operation of control devices, such as pumps and valves, adapts to consumer water use. Further development of the model could focus on simulating intermittent water supply systems and conditions where consumers are equipped with domestic storage tanks or pumps, which may provide valuable insights for future research. Computational challenges associated with scaling the model to larger populations and more intricate WDN should also be addressed, with strategies developed to ensure numerical stability and convergence in simulations. The model could also be applied in future studies to assess the enforceability of policies designed to encourage water-saving behaviors under conditions of reduced WDN pressure. Furthermore, the proposed framework could be extended to incorporate a wider range of consumer responses under water scarcity conditions, including water-conservation practices such as the use of efficient appliances and reducing the frequency of activities like bathing. Finally, extending the current model into a decision-support tool for practical use will require further steps for calibration and validation. These include assigning realistic consumer characteristics, refining the representation of water conservation attitudes and behaviors, and adjusting demand components based on empirical data. Integrating real-time monitoring data and conducting sensitivity analyses on key parameters are also recommended to support the practical applicability of the framework.

6. Conclusion

In the present study, an integrated modeling framework was developed to explicitly couple consumer behavior with the hydraulic conditions of the WDN. This framework addressed a key limitation of traditional approaches by capturing the dynamic interplay between individual water use behaviors and WDN hydraulic performance at fine temporal scales and across specific water use activities. By incorporating tap-level supply conditions, disaggregated demand profiles, and heterogeneity conservation attitudes, a more realistic simulation of water use was achieved. This framework was implemented in the WDN of Gaz City to evaluate its capability in modeling WDN hydraulic performance under different operating conditions. To assess the sensitivity of the proposed model to various WDN operating conditions, WDN hydraulic performance was analyzed under different reservoir head values. Results analysis indicated that at a reservoir head of 1625 m, although the WDN demand satisfaction ratio exceeding one, many consumers were unable to perform their activities as usual due to pressure drops. In one household, for instance, only 45 % of the requested water was received. Increasing the reservoir head to 1630 m improved the demand satisfaction ratio by approximately 10 % and reduced consumer dissatisfaction by nearly 97 %. At 1635 m, consumer satisfaction was fully achieved, with no further improvements observed at higher reservoir heads. The findings demonstrated that the present framework provides a more realistic assessment of WDN hydraulic performance compared to conventional hydraulic models by revealing service inequities that aggregated hydraulic indicators tend to obscure. Furthermore, it supports managers in identifying vulnerable areas and evaluating the social consequences of hydraulic decisions. The results also showed that the model captures how changes in consumer attitude influence water use and WDN hydraulic performance. Specifically, the findings indicated that at a fixed reservoir head, enhancing water conservation attitudes could alter the operating conditions of the WDN and reduce consumer water use by up to 13 %. This highlights the model's potential to assess demand management strategies and to inform more effective planning and implementation of conservation initiatives.

Statements & declarations

None.

Financial Disclosure Statement

The authors received no specific funding for this work.

Consent for Publication

All authors approved the submission and consented to publication.

Ethics Approval

Not applicable.

Consent to Participate

Not applicable.

CRedit authorship contribution statement

Azadeh Ahmadi: Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Zohre Fakhari:** Writing – original draft, Visualization, Validation, Software, Methodology, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

In this study, the social impact theory was employed in accordance with the methodology presented by Koutiva and Makropoulos (2016), under the assumption of no demand management policies, to simulate the effect of consumer interactions on their NAR. To model social impact, the discrete opinion dynamics method was used. Accordingly, consumer water conservation attitudes were defined in a binary form (positive or negative opinion). To convert the continuous water conservation attitude rank of consumers into a discrete form, the median of the attitude rank range was assumed as the boundary between positive and negative water conservation opinion. In implementing this theory, consumers with an NAR equal to or greater than 0.5 were considered to have a positive water conservation opinion, while those with an NAR below 0.5 were deemed to have a negative water conservation opinion. An individual with a positive water conservation opinion engaged in behaviors promoting water-saving practices, whereas a consumer with a negative water conservation opinion did not exhibit such behaviors. The social influence on consumer water conservation opinion was calculated using Eq. (B. 1).

$$SI_i = -S_i b - \sum_{ij} \frac{S_j O_j O_i}{d_{ij}^2} \quad (B1)$$

where SI_i is the exerted social impact on consumer i who is in interaction with n other consumers of covered WDN population, O_i indicates the consumer water conserving opinion, represented by values of -1 or $+1$, S_i is influence strength of each consumer (≥ 0). In the study by Koutiva and Makropoulos (2016), sensitivity analysis conducted during calibration process established the parameter S within the range of 2.199–3.647. Accordingly, a fixed influence strength value of 2.5 was assigned to all consumers in this study. b represents consumer stubbornness and was fixed at a value of 2. Each consumer j interacting with consumer i , was characterized by an influence strength S_j , an opinion O_j and a social distance d_{ij} . Social distance (d_{ij}) is calculated based on the sum of differences in properties between two connected consumers using Eq. (B. 2)

$$d_{ij} = 1 + \left| \sum_{i,j,n=1}^n x_{ni} - x_{nj} \right| \quad (B2)$$

where x_{ni} and x_{nj} represent the n -th property of consumers i and j , respectively, among N total properties, and x equal to grade of individual attributes such as age and education level, as well as grade of household attributes such as household size, income, dwelling size, ownership status (owner or tenant), and whether the water meter is shared in apartment buildings.

Then the probability that water conservation opinion of consumer i will change due to social impact SI_i calculated based on the concept of the volatility of social impact (Bahr and Passerini, 1998; Kacperski, 1999), adapted from Koutiva and Makropoulos (2016), using Eq. (B. 3).

$$P_{change} = \frac{\exp\left(\frac{SI_i}{T}\right)}{\exp\left(\frac{-SI_i}{T}\right) + \exp\left(\frac{SI_i}{T}\right)}$$

$$\begin{aligned} X &> P_{change} \rightarrow \text{No change in opinion} \\ X &< P_{change} \rightarrow \text{Change in opinion} \end{aligned} \quad (B3)$$

where P_{change} is the probability of a consumer changing their opinion, SI_i is the exerted social impact, and X is a randomly selected number from a uniform distribution of $[0, 1]$. T is the average volatility or social temperature of social impact ($T \geq 0$). In the study by Koutiva and Makropoulos (2016), sensitivity analysis estimated the average value of parameter T at approximately 10.5. Accordingly, in the present study, T was assumed to be 10.5. In this study, when a consumer's opinion was influenced through social interactions, it could shift from positive to negative or vice versa. If the opinion shifted from positive to negative, the grades associated with cognitive beliefs were randomly decreased; conversely, if the opinion shifted from negative to positive, these grades were randomly increased. The consumer's NAR was then recalculated.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ejrh.2025.102805](https://doi.org/10.1016/j.ejrh.2025.102805).

Data availability

Data will be made available on request.

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